

# ORF 524 — Midterm Exam

Matias D. Cattaneo

Department of Operations Research and Financial Engineering  
Princeton University

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### Notes:

- Duration of examination: 90 minutes.  
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).  
Please provide as much detail as possible in your answers.  
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.  
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.  
This is otherwise a closed-book and individual examination.

# 1 Question 1: Sufficiency and Completeness (25 points)

Let  $X \sim \mathbb{P}_\theta$  where  $\theta \in \Theta$ . Let  $S(X)$  be a statistic and  $m$  be a measurable function.

1. Show that if  $m(S(X))$  is a sufficient statistic then  $S(X)$  is a sufficient statistic.

**Solution:** Using the factorization theorem for  $m(S(X))$  gives

$$f(X, \theta) = h(X)g(m(S(X)), \theta) = h(X)q(S(X), \theta),$$

for measurable functions  $h$  and  $g$ , where  $q(s, \theta) = g(m(s), \theta)$ . So  $S(X)$  is sufficient also.

2. Show that if  $S(X)$  is sufficient and the inverse of  $m$  exists and is measurable then  $m(S(X))$  is sufficient.

**Solution:** Apply the previous part with  $S(X) = m^{-1}(m(S(X)))$ . Alternatively for a more direct proof, using the factorization theorem for  $S(X)$  gives

$$f(X, \theta) = h(X)g(S(X), \theta) = h(X)q(m(S(X)), \theta)$$

for measurable functions  $h$  and  $g$ , where  $q(s, \theta) = g(m^{-1}(s), \theta)$ .

3. Give an example of where both  $S(X)$  and  $m(S(X))$  are sufficient but  $m$  has no inverse.

**Solution:** Let  $X_i$  be i.i.d.  $\text{Uniform}(0, \theta)$  where  $\theta > 0$ . Take  $S(X) = X$  and  $m(S(X)) = \max_{1 \leq i \leq n} X_i$ . Many other examples are available.

4. Show that if  $S(X)$  is complete then  $m(S(X))$  is complete.

**Solution:** Take a measurable function  $g$  and define  $h = g \circ m$ . Suppose that

$$0 = \mathbb{E}_\theta[g(m(S(X)))] = \mathbb{E}_\theta[h(S(X))]$$

for all  $\theta$ . Then by completeness of  $S(X)$ ,

$$1 = \mathbb{P}_\theta[h(S(X)) = 0] = \mathbb{P}_\theta[g(m(S(X))) = 0]$$

for all  $\theta$  and so  $m(S(X))$  is also complete.

5. Supposing a minimal sufficient statistic exists, show that if  $S(X)$  is complete sufficient and bounded then  $S(X)$  is minimal sufficient.

**Solution:** Let  $T(X)$  be minimal sufficient, which implies that  $T(X) = r(S(X))$  for some measurable function  $r$  because  $S(X)$  is sufficient. We need to show  $S(X) = m(T(X))$  for a measurable function  $m$ . By boundedness define  $m(T(X)) = \mathbb{E}_\theta[S(X) \mid T(X)]$ , which is not a function of  $\theta$  by sufficiency of  $T(X)$ . Let  $g(s) = s - m(r(s))$  so that  $\mathbb{E}_\theta[g(S(X))] = \mathbb{E}_\theta[S(X) - m(r(S(X)))] = \mathbb{E}_\theta[S(X) - m(T(X))] = 0$  for all  $\theta$ , and by completeness of  $S(X)$

it follows that  $\mathbb{P}_\theta[g(S(X)) = 0] = 1$  for all  $\theta \in \Theta$ . Thus  $S(X) = m(r(S(X))) = m(T(X))$  almost surely. This result is known as *Bahadur's Theorem*.

## 2 Question 2: Finite-Sample Statistical Inference (40 points)

Let  $X = (X_i : 1 \leq i \leq n)$  where  $X_i \sim \mathbb{P}_\theta$  i.i.d. with Lebesgue density

$$f(x; \theta) = \frac{2x}{\theta^4} \mathbb{1}\{0 \leq x \leq \theta^2\}$$

for  $\theta > 0$ , where  $\mathbb{1}$  is the indicator function.

1. Find  $F(x; \theta) = \mathbb{P}_\theta(X_i \leq x)$ , the distribution function of  $X_i$ .

**Solution:** For  $x \in [0, \theta^2]$ ,

$$\int_0^x f(t; \theta) dt = \frac{2}{\theta^4} \int_0^x t dt = \frac{x^2}{\theta^4}.$$

Therefore

$$F(x; \theta) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{x^2}{\theta^4} & \text{if } 0 < x \leq \theta^2 \\ 1 & \text{if } \theta^2 < x \end{cases}.$$

2. Consider the following estimators:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \sqrt{X_i} \quad \text{and} \quad \hat{\theta}_{\text{ML}} = \max_{1 \leq i \leq n} \sqrt{X_i}.$$

- (a) Propose an unbiased estimator of  $\theta$  based on  $\hat{\theta}$  and denote it by  $\hat{\theta}_{\text{U}}$ . Can  $\hat{\theta}_{\text{U}}$  be interpreted as a method of moments estimator?

**Solution:**

$$\mathbb{E}_\theta[\hat{\theta}] = \mathbb{E}_\theta[\sqrt{X_i}] = \int_0^{\theta^2} \frac{2x}{\theta^4} \sqrt{x} dx = \frac{2}{\theta^4} \int_0^{\theta^2} x^{3/2} dx = \frac{2}{\theta^4} \frac{\theta^5}{5/2} = \frac{4}{5} \theta.$$

It follows that

$$\hat{\theta}_{\text{U}} = \frac{5}{4} \hat{\theta} = \frac{5}{4n} \sum_{i=1}^n \sqrt{X_i}$$

is an unbiased estimator of  $\theta$  based on  $\hat{\theta}$ . Further,  $\hat{\theta}_{\text{U}}$  is a method of moments estimator for the moment function  $g(u) = \sqrt{u}$  because  $\widehat{\mathbb{E}}_\theta[g(X_i)] = \frac{1}{n} \sum_{i=1}^n g(X_i) = \hat{\theta}_{\text{U}}/5$ , which is the sample analog of  $\mathbb{E}_\theta[g(X_i)] = 4\theta/5$ .

- (b) Show that  $\hat{\theta}_{\text{ML}}$  is the maximum likelihood estimator for  $\theta$  and that it is a complete sufficient statistic.

**Solution:** Firstly, the likelihood is the random variable

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta) = \frac{2^n}{\theta^{4n}} \left( \prod_{i=1}^n X_i \right) \mathbb{1}\left\{ \max_{1 \leq i \leq n} \sqrt{X_i} \leq \theta \right\},$$

so  $\hat{\theta}_{\text{ML}} = \max_{1 \leq i \leq n} \sqrt{X_i}$  is the MLE. Secondly,  $\max_{1 \leq i \leq n} \sqrt{X_i}$  is sufficient by the factorization theorem with

$$h(X) = \prod_{i=1}^n X_i \quad \text{and} \quad g\left(\max_{1 \leq i \leq n} \sqrt{X_i}, \theta\right) = \frac{2^n}{\theta^{4n}} \mathbb{1}\left\{\max_{1 \leq i \leq n} \sqrt{X_i} \leq \theta\right\}.$$

Thirdly, letting  $F_{\text{ML}}(x; \theta) = \mathbb{P}_{\theta}(\hat{\theta}_{\text{ML}} \leq x)$ , we have for  $x \geq 0$

$$F_{\text{ML}}(x; \theta) = \mathbb{P}_{\theta}\left(\max_{1 \leq i \leq n} X_i \leq x^2\right) = F(x^2; \theta)^n$$

by i.i.d. and therefore

$$F_{\text{ML}}(x; \theta) = \begin{cases} 0 & \text{if } x \leq 0 \\ \left(\frac{x}{\theta}\right)^{4n} & \text{if } 0 < x \leq \theta \\ 1 & \text{if } \theta < x \end{cases}.$$

Let  $g$  be a measurable function satisfying

$$0 = \mathbb{E}[g(\hat{\theta}_{\text{ML}})] = \int_0^{\theta} g(x) f_{\text{ML}}(x; \theta) dx = \int_0^{\theta} g(x) \frac{4nx^{4n-1}}{\theta^{4n}} dx$$

for all  $\theta$ . Since  $\{[0, \theta] : \theta > 0\}$  generates the Borel  $\sigma$ -algebra on  $[0, \infty)$ , we must have

$$\int_A g(x) x^{4n-1} dx = 0$$

for all Borel sets  $A \subseteq [0, \infty)$ . Now suppose that  $g$  is not zero Lebesgue-almost everywhere. Then without loss of generality we have that  $\text{Leb}\{g \geq \varepsilon\} \geq \varepsilon$  for some  $\varepsilon > 0$ . But then

$$\int_{\{g \geq \varepsilon\}} g(x) x^{4n-1} dx \geq \int_{\{g \geq \varepsilon\} \cap \{x \geq \varepsilon/2\}} g(x) x^{4n-1} dx \geq \frac{\varepsilon}{2} \varepsilon \left(\frac{\varepsilon}{2}\right)^{4n-1} > 0$$

is a contradiction. So  $g = 0$  Lebesgue-almost everywhere and as the law of  $\hat{\theta}_{\text{ML}}$  is absolutely continuous with respect to the Lebesgue measure, we also have  $g(\hat{\theta}_{\text{ML}}) = 0$ ,  $\mathbb{P}$ -almost surely. So  $\hat{\theta}_{\text{ML}}$  is complete.

- (c) Find a uniformly minimum variance unbiased (UMVU) estimator  $\hat{\theta}_{\text{UMVU}}$  for  $\theta$ .

**Solution:** We find an unbiased estimator based on the complete sufficient statistic  $\hat{\theta}_{\text{ML}}$  and apply the Lehmann–Scheffé theorem. We have

$$f_{\text{ML}}(x; \theta) = \frac{4nx^{4n-1}}{\theta^{4n}} \mathbb{1}\{0 \leq x \leq \theta\}$$

and

$$\mathbb{E}_{\theta}[\hat{\theta}_{\text{ML}}] = \int_0^{\theta} x f_{\text{ML}}(x; \theta) dx = \int_0^{\theta} \frac{4nx^{4n}}{\theta^{4n}} dx = \frac{4n\theta^{4n+1}}{\theta^{4n}(4n+1)} = \frac{4n}{4n+1}\theta.$$

Therefore by linearity of expectation and the Lehmann–Scheffé theorem a UMVU estimator for  $\theta$  is

$$\hat{\theta}_{\text{UMVU}} = \frac{4n+1}{4n} \hat{\theta}_{\text{ML}}.$$

- (d) Calculate  $\mathbb{V}_\theta[\hat{\theta}_{\text{U}}]$  and  $\mathbb{V}_\theta[\hat{\theta}_{\text{UMVU}}]$  and give a necessary and sufficient condition under which  $\mathbb{V}_\theta[\hat{\theta}_{\text{U}}] > \mathbb{V}_\theta[\hat{\theta}_{\text{UMVU}}]$ .

**Solution:** We have

$$\begin{aligned} \mathbb{V}_\theta[\hat{\theta}_{\text{U}}] &= \frac{25}{16n} \mathbb{V}_\theta[\sqrt{X_i}] = \frac{25}{16n} \left( \mathbb{E}_\theta[X_i] - \left( \mathbb{E}_\theta[\sqrt{X_i}] \right)^2 \right) = \frac{25}{16n} \left( \frac{2\theta^2}{3} - \left( \frac{4\theta}{5} \right)^2 \right) \\ &= \frac{25\theta^2}{16n} \frac{2}{75} = \frac{\theta^2}{24n} \end{aligned}$$

and

$$\begin{aligned} \mathbb{V}_\theta[\hat{\theta}_{\text{UMVU}}] &= \frac{(4n+1)^2}{16n^2} \mathbb{V}[\hat{\theta}_{\text{ML}}] = \frac{(4n+1)^2}{16n^2} \left( \mathbb{E}[\hat{\theta}_{\text{ML}}^2] - \mathbb{E}[\hat{\theta}_{\text{ML}}]^2 \right) \\ &= \frac{(4n+1)^2}{16n^2} \left( \int_0^\theta \frac{4nx^{4n+1}}{\theta^{4n}} dx - \frac{16n^2}{(4n+1)^2} \theta^2 \right) \\ &= \frac{(4n+1)^2}{16n^2} \left( \frac{4n\theta^{4n+2}}{\theta^{4n}(4n+2)} - \frac{16n^2}{(4n+1)^2} \theta^2 \right) \\ &= \frac{(4n+1)^2}{16n^2} \left( \frac{4n}{(4n+2)} \theta^2 - \frac{16n^2}{(4n+1)^2} \theta^2 \right) \\ &= \frac{\theta^2}{16n^2} \left( \frac{4n(4n+1)^2 - 16n^2(4n+2)}{(4n+2)} \right) \\ &= \frac{\theta^2}{16n^2 + 8n}, \end{aligned}$$

which shows that  $\mathbb{V}_\theta[\hat{\theta}_{\text{UMVU}}] = \mathbb{V}_\theta[\hat{\theta}_{\text{U}}]$  for  $n = 1$  (because then  $\hat{\theta}_{\text{UMVU}} = \hat{\theta}_{\text{U}}$ ), but otherwise  $\mathbb{V}_\theta[\hat{\theta}_{\text{UMVU}}] < \mathbb{V}_\theta[\hat{\theta}_{\text{U}}]$  for  $n > 1$ .

3. Consider the following hypothesis testing problem:

$$H_0 : \theta \leq \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0$$

where  $\theta_0 > 0$ . We use a test that rejects  $H_0$  if and only if  $\hat{\theta}_{\text{ML}}/\theta_0 > c$  for some critical value  $c$ .

- (a) Find  $c$  so that the test has 5% size.

**Solution:**

$$\begin{aligned} \mathbb{P}_\theta(\hat{\theta}_{\text{ML}}/\theta_0 > c) &= 1 - \mathbb{P}_\theta(\hat{\theta}_{\text{ML}}/\theta_0 \leq c) = 1 - F_{\text{ML}}(c\theta_0; \theta) \\ &= \begin{cases} 1 - (c\theta_0/\theta)^{4n} & \text{if } 0 \leq c < \theta/\theta_0 \\ 0 & \text{if } c \geq \theta/\theta_0 \end{cases} \end{aligned}$$

$$= \max\{1 - (c\theta_0/\theta)^{4n}, 0\}$$

for any  $c > 0$  and any  $\theta > 0$ . Since  $\mathbb{P}_\theta(\hat{\theta}_{\text{ML}}/\theta_0 > c)$  is non-decreasing in  $\theta$ ,

$$\sup_{\theta \leq \theta_0} \mathbb{P}_\theta(\hat{\theta}_{\text{ML}}/\theta_0 > c) = \mathbb{P}_{\theta_0}(\hat{\theta}_{\text{ML}}/\theta_0 > c) = \max\{1 - c^{4n}, 0\},$$

and therefore  $c = (0.95)^{1/(4n)}$ .

(b) Derive and sketch a plot of the power function  $\beta(\theta)$ .

**Solution:** We have

$$\begin{aligned} \beta(\theta) &= \mathbb{P}_\theta(\hat{\theta}_{\text{ML}}/\theta_0 > (0.95)^{1/(4n)}) = \max\{1 - ((0.95)^{1/(4n)}\theta_0/\theta)^{4n}, 0\} \\ &= \max\{1 - 0.95(\theta_0/\theta)^{4n}, 0\}. \end{aligned}$$

(c) Propose a 95% one-sided confidence interval for  $\theta$  based on this test.

**Solution:**  $[\hat{\theta}_{\text{ML}}/(0.95)^{1/(4n)}, \infty)$  is a 95% confidence interval because

$$\begin{aligned} \mathbb{P}_\theta \left( \theta \in [\hat{\theta}_{\text{ML}}/(0.95)^{1/(4n)}, \infty) \right) &= \mathbb{P}_\theta \left( \hat{\theta}_{\text{ML}}/(0.95)^{1/(4n)} \leq \theta \right) \\ &= \mathbb{P}_\theta \left( \hat{\theta}_{\text{ML}} \leq (0.95)^{1/(4n)}\theta \right) = 0.95. \end{aligned}$$

